

CDKN2B fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes CDKN2B gene fusions from the msk_impact_50k_2026 study, covering 12 fusion events. 2/12 fusions (16.7%) were in-frame. 2/12 fusions (16.7%) retained the Ankyrin repeats (3 copies) domain. Domain retention was tested with Fisher's exact test ($p=0.0222222$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for CDKN2B.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 12 analyzed fusion events, identifying 7 distinct fusion partner genes. The most frequent partner was CDKN2A, observed in 6 events.

Domain retention

Ankyrin repeats (3 copies) domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 2/2 (100.0%) of in-frame fusions retained the domain, $p=0.0222222$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

Domain disruption

Domain-disruption analysis was skipped: CDKN2B has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 10 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 52 aa (permutation-corrected $p=0.762376$, not statistically significant at $\alpha=0.05$). This is -2 aa from the nearest configured domain boundary at 54 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=2$) and "not_retained" ($n=8$). For Frame_status, retained: 2/2 (100.0%, 95% CI 34.2%-100.0%); not_retained: 0/1 (0.0%, 95% CI 0.0%-79.3%). Welch's t-test comparing Tumor_variant_count between groups gave means of 11.5 vs 48.25, $p=0.170234$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: CDKN2B has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CDKN2A	6	0.5	0.20043213737826426	None	0.6741935483870968	0.2744647709419452	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.4108544296210313	1
CNTRL	1	0.08333333333333333	0.20043213737826426	None	1.0	0.2744647709419452	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.31569313929845066	2
MTAP	1	0.08333333333333333	0.20043213737826426	None	1.0	0.2744647709419452	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.31569313929845066	3
SLC24A2	1	0.08333333333333333	0.20043213737826426	None	0.7741935483870968	0.2744647709419452	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.27335442962103135	4
GAB2	1	0.08333333333333333	0.20043213737826426	None	0.6612903225806452	0.2744647709419452	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.25218507478232166	5
ELAVL2	1	0.08333333333333333	0.20043213737826426	None	None	0.2744647709419452	['confidence_certainty', 'domain_retention', 'recurrence']	0.15777617144424705	6
CDKN2B-AS1	1	0.08333333333333333	0.20043213737826426	None	0.0	0.2744647709419452	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.12819313929845072	7

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
38	0	2	1	7	0.0	1.0	-0.0
52	2	0	3	5	None	0.44444444444444453	0.3521825181113624
69	2	0	4	4	None	0.4666666666666668	0.3309932190414243
73	2	0	5	3	None	1.0	-0.0
75	2	0	6	2	None	1.0	-0.0
113	2	0	7	1	None	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

2	0
0	8

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Ankyrin repeats (3 copies)	10	2	5	3	0.34104938271604934

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
2	2	12	0.16666666666666666

frequency tables

Partner_gene_counts

Partner_gene	Event_count
CDKN2A	6
CDKN2B-AS1	1
CNTRL	1
ELAVL2	1
GAB2	1
MTAP	1
SLC24A2	1

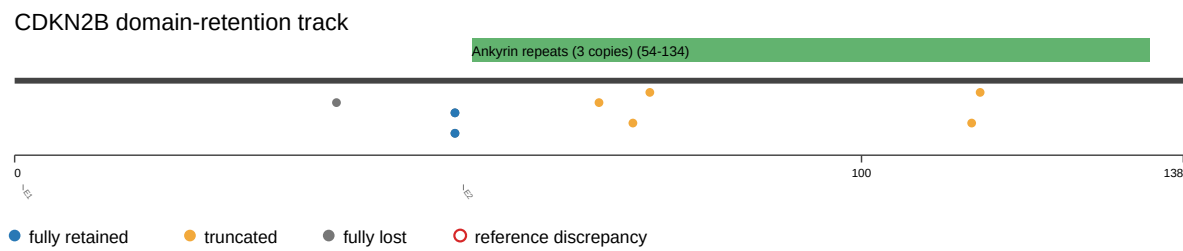
Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 5576-T01-I M6-2	P-0035 576-T0 1-IM6		CDKN2A -CDKN2 B	CDKN2A	unknown	five_prime	22006179	2	75	False	disrupted	0.2716049 38271604 9	True			Ankyrin repeats (3 copies)	CDKN2B (NM_004936) - CDKN2A (NM_000077) Rearrangement: c.224:CDKN2B_c.153:CDKN2A del Protein Fusion: mid-exon {CDKN2B:CDKN2A}	NA
EVT-P-001 0183-T02-I M6-10	P-0010 183-T0 2-IM6		CDKN2B -CDKN2 A	CDKN2A	unknown	five_prime	22006197	2	69	False	disrupted	0.1975308 64197530 85	True			Ankyrin repeats (3 copies)	CDKN2B (NM_004936) - CDKN2A (NM_058195) rearrangement: c.206:CDKN2B_c.194-11:CDKN2A del Protein Fusion: mid-exon {CDKN2B:CDKN2A}	NA
EVT-P-006 0581-T01-I M7-11	P-0060 581-T0 1-IM7		CDKN2B -CDKN2 A	CDKN2A	out-of-frame	five_prime	22008168	1	52	True	lost	0.0	False		Ankyrin repeats (3 copies)		CDKN2B (NM_004936) - CDKN2A (NM_000077) rearrangement: c.156+629:CDKN2B_c.458-143:CDKN2A del Protein Fusion: out of frame {CDKN2B:CDKN2A}	NA
EVT-P-004 5019-T02-I M6-12	P-0045 019-T0 2-IM6		CDKN2B -CDKN2 A	CDKN2A	unknown	five_prime	22006065	2	113	False	disrupted	0.7407407 40740740 7	True			Ankyrin repeats (3 copies)	CDKN2B (NM_004936) - CDKN2A (NM_058195) rearrangement: c.338:CDKN2B_c.303:CDKN2A del Protein Fusion: mid-exon {CDKN2B:CDKN2A}	NA
EVT-P-003 8263-T01-I M6-13	P-0038 263-T0 1-IM6		CDKN2B -CDKN2 A	CDKN2A	unknown	five_prime	22007679	1	52	True	lost	0.0	False		Ankyrin repeats (3 copies)		CDKN2B (NM_004936) - CDKN2A (NM_000077) Rearrangement: c.156+1118:CDKN2B_c.192:CDKN2A del Protein Fusion: mid-exon {CDKN2B:CDKN2A}	NA
EVT-P-000 9347-T01-I M5-34	P-0009 347-T0 1-IM5		CDKN2B -CDKN2 B-AS1	CDKN2B-AS1	unknown	five_prime	22006062	2	114	False	disrupted	0.7530864 19753086 4	True			Ankyrin repeats (3 copies)	CDKN2B (NM_004936) rearrangement : c.341_chr9:g.22015263inv Protein fusion: mid-exon (CDKN2B-CDKN2BAS)	NA
EVT-P-004 5616-T01-I M6-36	P-0045 616-T0 1-IM6		CDKN2B -SLC24A 2	SLC24A2	unknown	five_prime	22008840	1	38	False	lost	0.0	False		Ankyrin repeats (3 copies)		CDKN2B (NM_004936) - SLC24A2 (NM_020344) rearrangement: c.113:CDKN2B_c.931-51322:SLC24A2 del Protein Fusion: mid-exon {CDKN2B:SLC24A2}	NA

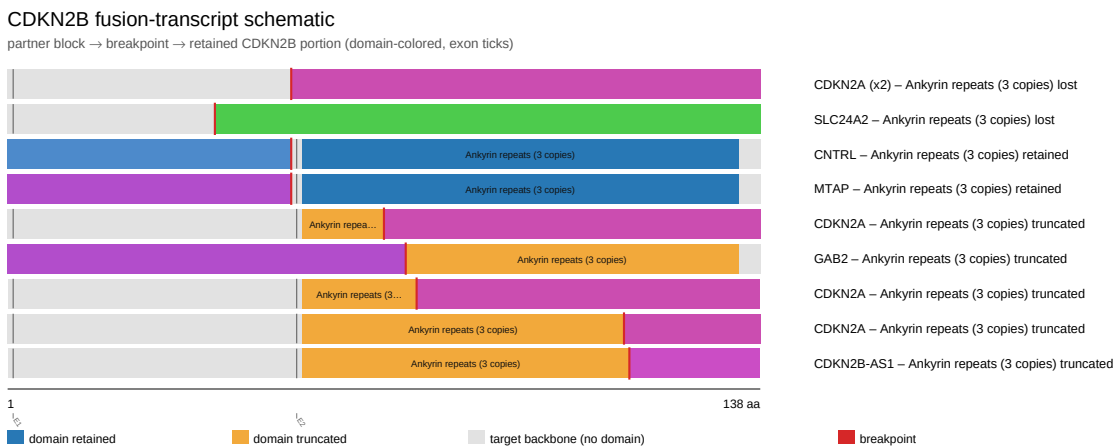
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 2985-T01-IM6-37	P-0042 985-T01-IM6		CNTRL- CDKN2B	CNTRL	in-frame	three_prime	22008250	1	52	True	retained	1.0	False	Ankyrin repeats (3 copies)			CNTRL (NM_007018) - CDKN2B (NM_004936) rearrangement: c.1455+1114:CNTRL_c.156+547:CDKN2Binv Protein Fusion: in frame {CNTRL:CDKN2B}	NA
EVT-P-005 3040-T01-IM6-38	P-0053 040-T01-IM6		GAB2- CDKN2B	GAB2	unknown	three_prime	22006184	2	73	False	disrupted	0.7654320987654321	True			Ankyrin repeats (3 copies)	GAB2 (NM_080491) - CDKN2B (NM_004936) rearrangement: t(9;11)(p21.3;q14.1)(chr9:g.22006184::chr11:g.78068634) Protein Fusion: mid-exon {GAB2:CDKN2B}	NA
EVT-P-000 2886-T01-IM3-40	P-0002 886-T01-IM3		MTAP- CDKN2B	MTAP	in-frame	three_prime	22008292	1	52	True	retained	1.0	False	Ankyrin repeats (3 copies)			NA Protein fusion: in frame (MTAP-CDKN2B)	NA

Figures

domain retention outliers



fusion schematic



reference comparison

Reference vs msk_impact_50k_2026

